

Potential role of FoxO1 and mTORC1 in the pathogenesis of Western diet-induced acne

Acne in adolescents of developed countries is an epidemic skin disease and has currently been linked to the Western diet (WD). It is the intention of this viewpoint to discuss the possible impact of WD-mediated nutrient signalling in the pathogenesis of acne. High glycaemic load and dairy protein consumption both increase insulin/insulin-like growth factor-1 (IGF-1) signalling (IIS) that is superimposed on elevated IGF-1 signalling of puberty. The cell's nutritional status is primarily sensed by the forkhead box transcription factor O1 (FoxO1) and the serine/threonine kinase mammalian target of rapamycin complex 1 (mTORC1). Increased IIS extrudes FoxO1 into the cytoplasm, whereas nuclear FoxO1 suppresses hepatic IGF-1 synthesis and thus impairs somatic growth. FoxO1 attenuates androgen signalling, interacts with regulatory proteins important for sebaceous lipogenesis, regulates the activity of innate and adaptive immunity, antagonizes oxidative stress and most importantly functions as a rheostat of mTORC1, the master regulator of cell growth, proliferation and metabolic homeostasis. Thus, FoxO1 links nutrient availability to mTORC1-driven processes: increased protein and lipid synthesis, cell proliferation, cell differentiation including hyperproliferation of acroinfundibular keratinocytes, sebaceous gland hyperplasia, increased sebaceous lipogenesis, insulin resistance and increased body mass index. Enhanced androgen, TNF- α and IGF-1 signalling due to genetic polymorphisms promoting the risk of acne all converge in mTORC1 activation, which is further enhanced by nutrient signalling of WD. Deeper insights into the molecular interplay of FoxO1/mTORC1-mediated nutrient signalling are thus of critical importance to understand the impact of WD on the promotion of epidemic acne and more serious mTORC1-driven diseases of civilization.